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Chapter 5

Strategic Capacity Planning

Operations Management

FOURTEENTH EDITION

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Learning Objective: Chapter 5

You should be able to:

- Explain the importance of *capacity planning*
- Describe ways of defining and *measuring capacity*
- Name several determinants of *effective capacity*
- Discuss the major considerations related to *developing capacity alternatives*
- Describe approaches that are useful for *evaluating capacity alternatives*

Capacity Planning

- **Capacity** refers to an upper limit or *ceiling on the load* that an operating unit can handle.

Operating unit	Capacity needs
• Plant • Department • Machine • Store • Worker	• Equipment • Space • Employee skills

LO 5.1

Strategic Capacity Planning

- The goal of **strategic capacity planning** is to achieve a match between *the long-term supply capabilities* of an organization and *the predicted level of long-term demand*.

Overcapacity

- operating costs that are too high

Undercapacity

- Strained resources and possible loss of customers

LO 5.1

Capacity Planning Questions

- **Key questions:**
  - **What kind** of capacity is needed?
  - **How much** is needed to match demand?
  - **When** is it needed?
- **Related questions:**
  - How much will it **cost**?
  - What are the **potential benefits** and **risks**?
  - Are there **sustainability** issues?
  - Can **supply chain** handle the necessary changes of capacity?

LO 5.1

Capacity Decisions are Strategic

- **Capacity decisions** are strategic and critical for an organization, the reasons including:
  1. Impact the ability of the organization to meet future demands
  2. Affect operating costs
  3. Are a major determinant of initial cost
  4. Often involve long-term commitment of resources
  5. Can affect competitiveness
  6. Affect the ease of management
  7. Have become more important and complex due to globalization
  8. Need to be planned for in advance due to their consumption of financial and other resources

LO 5.2

Defining and Measuring Capacity

- In selecting a *measure of capacity*, it is important to choose one that does not require updating.
- Two useful definitions of capacity:
  - **Design capacity**
    - The ***maximum output rate*** or service capacity an operation, process, or facility is designed for.
  - **Effective capacity**
    - Design capacity minus *allowances* such as personal time, maintenance or changing product mix.
    - It can be viewed as the ***economic output rate*** of a resource can achieve.

LO 5.3

Measures of capacity

Business	Inputs	Outputs
Auto manufacturing	Labor hours, machine hours	Number of cars per shift
Steel mill	Furnace size	Tons of steel per day
Oil refinery	Refinery size	Gallons of fuel per day
Farming	Number of acres, number of cows	Bushels of grain per acre per year, gallons of milk per day
Restaurant	Number of tables, seating capacity	Number of meals served per day
Theater	Number of seats	Number of tickets sold per performance
Retail sales	Square feet of floor space	Revenue generated per day

LO 5.3

Measuring System Effectiveness

- Different measures of capacity are useful in defining two measures of system effectiveness.
  - **Capacity Efficiency** is the ratio of actual output to effective capacity.

$$\text{Efficiency} = \frac{\text{Actual output}}{\text{Effective capacity}} \times 100\%$$

  - **Capacity Utilization** is the ratio of actual output to design capacity.

$$\text{Utilization} = \frac{\text{Actual output}}{\text{Design capacity}} \times 100\%$$

LO 5.3

Measuring System Effectiveness

- ***Practice:***

Given the following information, compute the efficiency and the utilization of the vehicle repair department:

- Design Capacity = 50 trucks per day
- Effective Capacity = 40 trucks per day
- Actual Output = 36 trucks per day

LO 5.3

Measuring System Effectiveness

- ***Solution:***

$$\text{Efficiency} = \frac{\text{Actual output}}{\text{Effective capacity}} \times 100\%$$

$$= \frac{36 \text{ trucks per day}}{40 \text{ trucks per day}} \times 100\% = 90\%$$

$$\text{Utilization} = \frac{\text{Actual output}}{\text{Design capacity}} \times 100\%$$

$$= \frac{36 \text{ trucks per day}}{50 \text{ trucks per day}} \times 100\% = 72\%$$

LO 5.3

Determinants of Effective Capacity

- Facilities factors
- Product and service factors
- Process factors
- Human factors
- Policy factors
- Operational factors
- Supply chain factors
- External factors

LO 5.4

Factors that determine effective capacity

A. Facilities

1. Design
2. Location
3. Layout
4. Environment

B. Product/service

1. Design
2. Product or service mix

C. Process

1. Quantity capabilities
2. Quality capabilities

D. Human factors

1. Job content
2. Job design
3. Training and experience
4. Motivation
5. Compensation
6. Learning rates
7. Absenteeism and labor turnover

E. Policy

F. Operational

1. Scheduling
2. Materials management
3. Quality assurance
4. Maintenance policies
5. Equipment breakdowns

G. Supply chain

H. External factors

1. Product standards
2. Safety regulations
3. Unions
4. Pollution control standards

LO 5.4

Formulation of Capacity Strategies

- **Capacity strategies** are typically based on *assumptions* and *predictions* about:

Long-term demand patterns	Technological change	Competitor behavior

LO 5.4

Primary Capacity Strategies

- Three primary capacity strategies:

Leading	• Build capacity in anticipation of future demand increases.
Following	• Build capacity when demand exceeds current capacity.
Tracking	• Adds capacity in relatively small increments to keep pace with increasing demand.

LO 5.4

Capacity Cushion

- A capacity cushion is extra capacity that is used to offset demand uncertainty.
  - *Capacity Cushion = Capacity - Expected Demand*
- Capacity cushion strategy
  - Organizations that have ***greater demand uncertainty*** typically have greater capacity cushions.
  - Organizations that have ***standard products and services*** generally have smaller capacity cushions.

LO 5.5

Steps in Capacity Planning

1. Estimate future capacity *requirements*
2. Evaluate existing capacity and facilities; *identify gaps*
3. Identify *alternatives* for meeting requirements
4. Conduct *financial analyses*
5. Assess key *qualitative issues*
6. Select the *best alternative* for the long-term
7. *Implement alternative* chosen
8. *Monitor* results

LO 5.5

Capacity planning decisions

- **Long-term considerations**
  - Relate to ***overall level*** of capacity requirements.
  - Require forecasting demand **over a *time horizon*** and converting those needs into capacity requirements.
  - Demand can exhibit a number of patterns such as a **trend** or **cycle**.
- **Short-term considerations**
  - Relate to ***probable variations*** in capacity requirements.
  - Less concerned with cycles and trends than with **random** and **irregular variations** from **average** and **deviations**.

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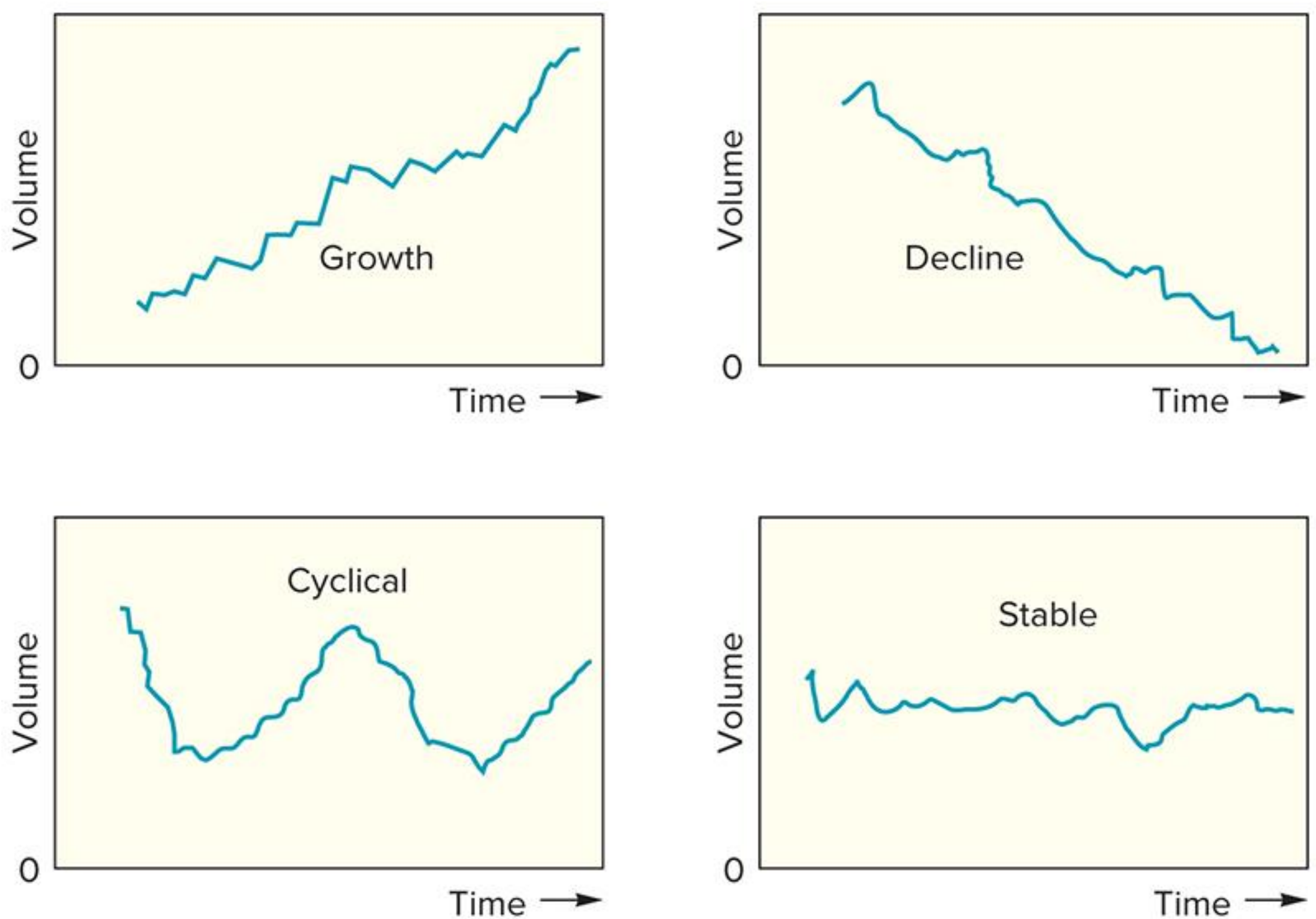
Common Demand Patterns



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Calculating Processing Requirements

- Calculating **processing requirements** requires reasonably accurate demand forecasts, standard processing times, and available work time.

- ***Equation:***

$$N_R = \frac{\sum_{i=1}^{k} p_i D_i}{T}$$

where

N_R = Units of capacity required

p_i = standard processing time for product i

D_i = demand for product i during the planning horizon

T = processing time available during the planning horizon

LO 5.6

Calculating Processing Requirements

- ***Practice:***

  A department works *one 8-hour shift*, *250 days a year*. Following table lists the information currently used by a machine. Please calculate **how many of the machines are needed per year**.

Product	Annual Demand	Standard Processing Time per Unit (hr)	Processing Time Needed (hr)
1	400	5.0	
2	300	8.0	
3	700	2.0	

Calculating Processing Requirements

- ***Solution:***

Product	Annual Demand	Standard Processing Time per Unit (hr)	Processing Time Needed (hr)
1	400	5.0	**2,000**
2	300	8.0	**2,400**
3	700	2.0	**1,400**
		Total of processing time needed =	**5,800**

Total processing time of a machine per year = 8 x 250 = 2,000

$$\text{Units of capacity requirements} = \frac{5{,}800 \text{ hours}}{2{,}000 \text{ hours/machine}} = 2.9$$

A: 3 of these machines would be needed.

In-House or Outsource?

- Once capacity requirements are determined, the organization must decide whether to produce a good or service *itself* or *outsource*.
- Factors to consider when deciding whether to *outsource*:
  - Available capacity
  - Expertise considerations
  - Quality considerations
  - Cost considerations
  - Risks considerations
  - The nature of demand

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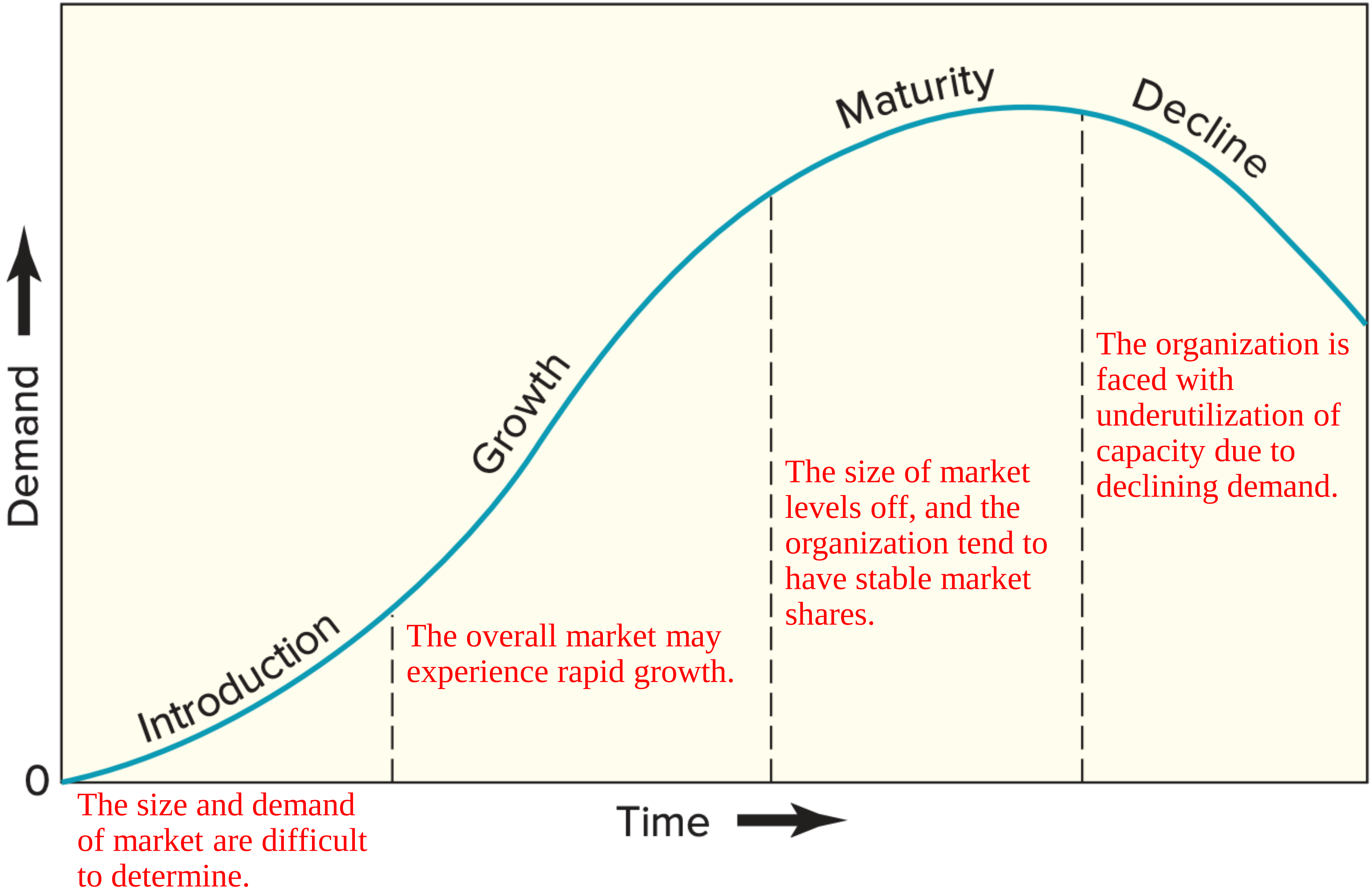
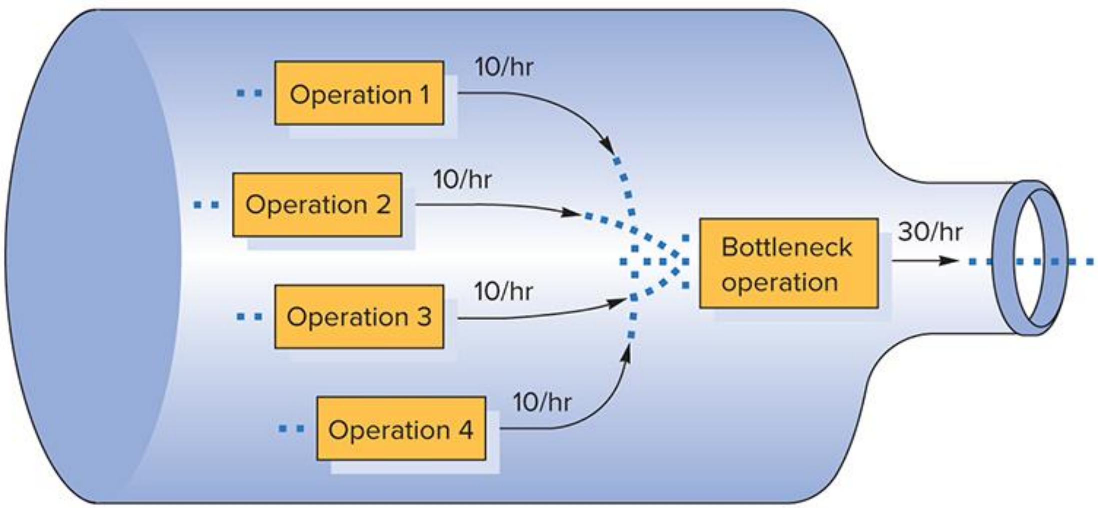
Considering Life Stage Cycle



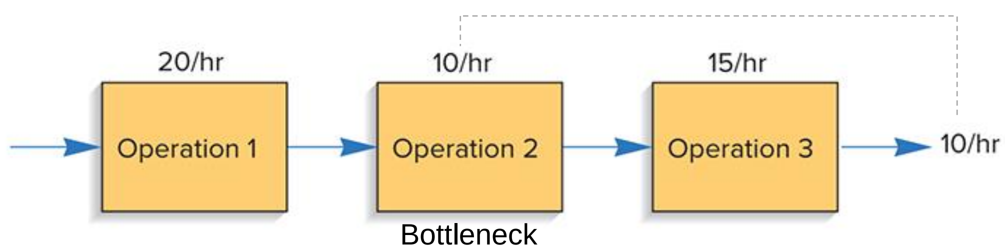
Bottleneck Operation

- **Bottleneck operation** is an operation in a sequence of operations whose capacity is lower than the capacities of other operations in the sequence.

Example 1:



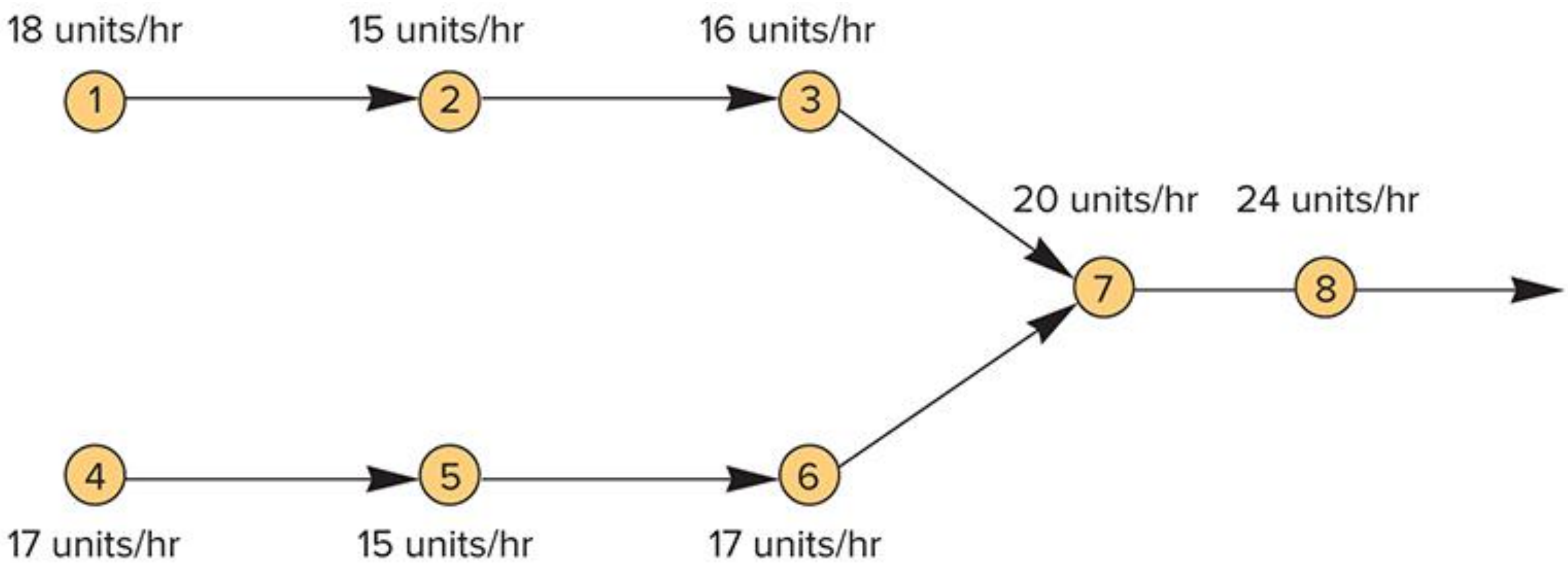
Example 2:



Bottleneck Operation

- ***Practice 1:***

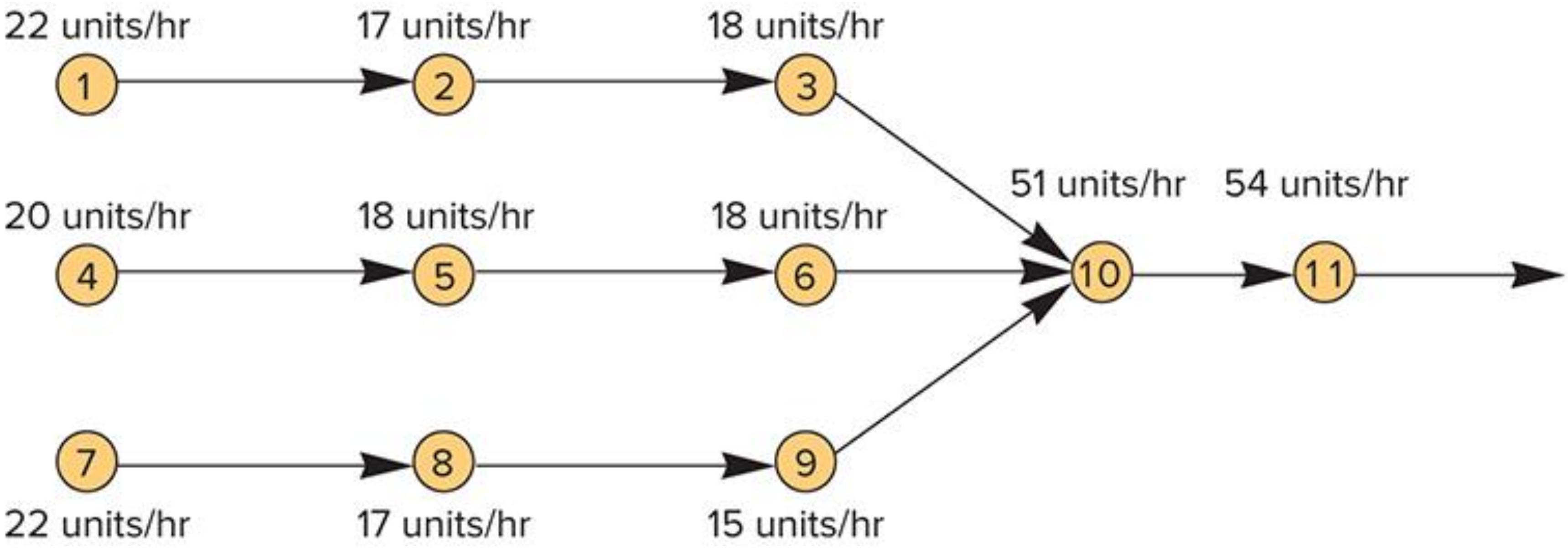
  What is the capacity on this system? (Circles represent operations.)



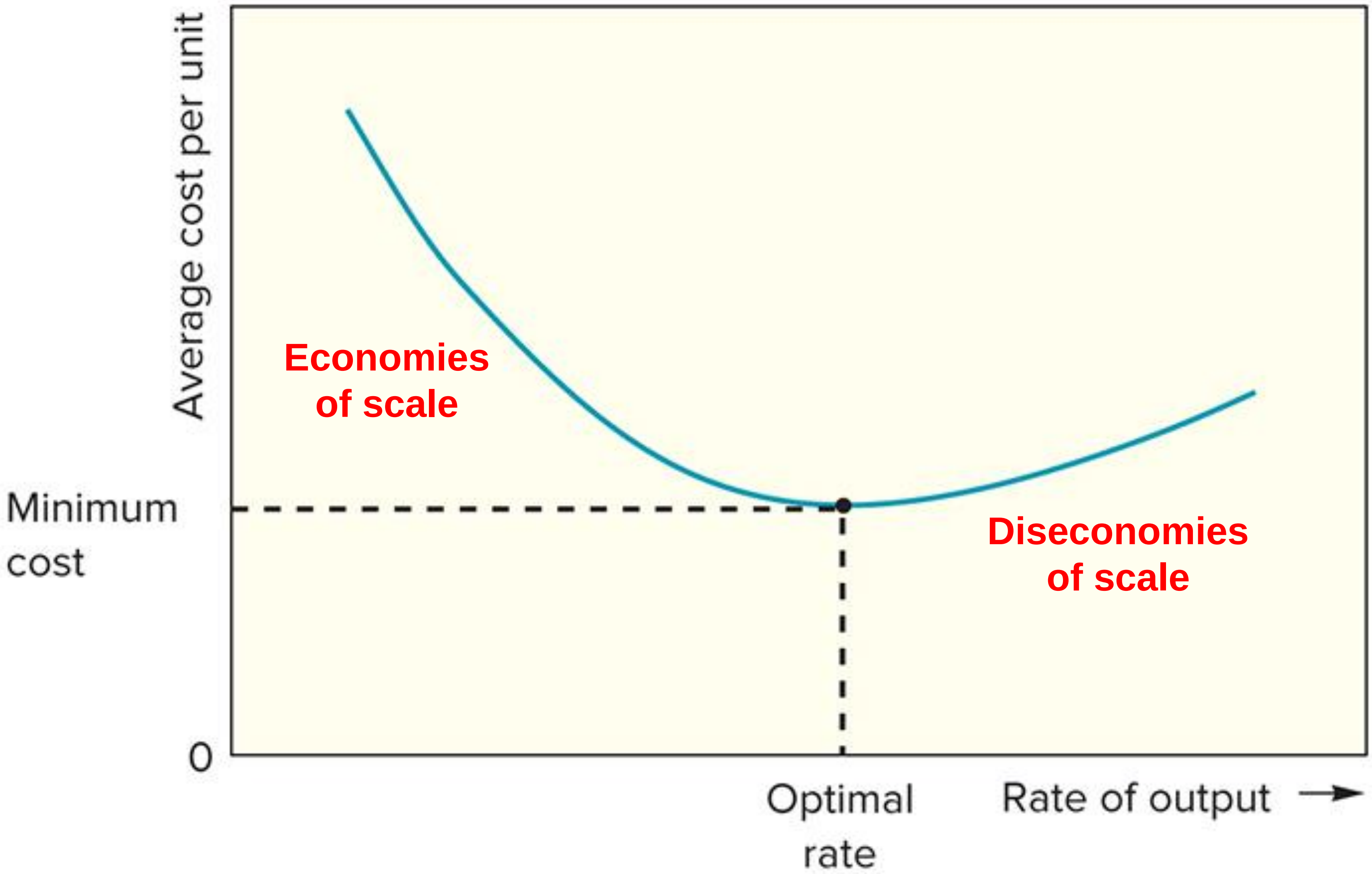
Bottleneck Operation

- ***Practice 2:***

  What is the capacity on this system?(Circles represent operations.)



Optimal Operating Level



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Economies and Diseconomies of Scale

Economies of scale

If output rate is less than the optimal level, increasing the output rate results in decreasing average per unit costs.

Diseconomies of scale

If the output rate is more than the optimal level, increasing the output rate results in increasing average costs per unit.

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Economies of Scale

- Reasons for economies of scale:

Fixed costs are spread over a larger number of units

Construction costs increase at a decreasing rate as facility size increases

Processing costs decrease due to standardization

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Diseconomies of Scale

- Reasons for diseconomies of scale

Distribution costs increase due to traffic congestion

Complexity increases costs

Inflexibility can be an issue

Additional levels of bureaucracy

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Evaluating Capacity Alternatives

- Capacity alternatives should be evaluated from varying perspectives.
- Typical questions about **economic consideration:**
  - Is it economically feasible?
  - How much will it cost?
  - How soon can we have it?
  - What will operating and maintenance costs be?
  - What will its useful life be?
  - Will it be compatible with present personnel and present operations?

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Evaluating Capacity Alternatives (cont.)

- Techniques for Evaluating Alternatives
  - **Cost-volume analysis**
  - **Financial analysis**
  - Waiting-line analysis
  - Simulation

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Cost-Volume Analysis

- **Cost-volume analysis** focuses on the relationship between ***cost***, ***revenue***, and ***volume of output***.
  - **Fixed Costs (FC)**
    - Tend to *remain constant* regardless of output volume
      - Ex: rental costs, property taxes, equipment costs, heating and cooling expenses, administrative costs
  - **Variable Costs (VC)**
    - Vary *directly with output volume*
      - Ex: material costs and labor costs

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Cost-Volume Analysis

- The input of **cost-volume analysis** revolves around **total cost** and **total revenue**.

- ***Equation:***

Total Cost (*TC*) = *FC* + *VC* = *FC* + v * *Q*

Where

v = Variable cost per unit

Q = Quantity of output

$VC = v * Q$

LO 5.11

Cost-Volume Analysis

- The input of **cost-volume analysis** revolves around **total cost** and **total revenue**.
- ***Equation:***

$$\textbf{Total Revenue } (TR) = R \times Q$$

Where

R = Revenue per unit

Q = Quantity of output

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Cost-Volume Analysis

- **Break-Even Point (BEP)**
  - Output volume at which total cost and total revenue are equal.
- ***Equations:***

$$\text{Profit } (P) = TR - TC = (R \times Q) - (FC + v * Q)$$

$$= Q(R - v) - FC$$

$$Q = \frac{P + FC}{R - v}$$

$$Q_{BEP} = \frac{FC}{R - v}$$

$*$ ***R* - *v*** is known as ***Contribution margin*.**

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Cost-Volume Analysis

- Summarization the symbols of cost–volume.

FC = Fixed cost
VC = Total variable cost
v = Variable cost per unit
TC = Total cost
TR = Total revenue
R = Revenue per unit
Q = Quantity or volume of output
Q_{BEP} = Break-even quantity
P = Profit

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Cost-Volume Analysis

- ***Practice:***

  An owner of Berry Pies is considering to add a new line of pies, which will require leasing new equipment for **a monthly payment of $6,000**. **Variable costs would be $2** per pie, and **pies would retail for $7** each.

  a) How many pies must be sold in order to break even?

  b) What would the profit (loss) be if 1,000 pies are made and sold in a month?

  c) How many pies must be sold to realize a profit of $4,000?

  d) If 2,000 pies can be sold, and a profit target is $5,000, what price should be charged per pie?

Cost-Volume Analysis

- ***Solution:***
  - Fixed cost(FC) = **$6,000**
  - Variable cost per unit (v) = **2 per pie**
  - Revenue per unit(R) = $**7 per pie**

a) ***BEP = FC / R - v*** = 6,000 / (7-2) = 1,200 (pies/month)

b) For Quality(Q) = 1,000;

Profit(P) = Q(R - v) - FC = 1,000 * (7-2) – 6,000 = -1,000

c) Profit(P) =4,000;

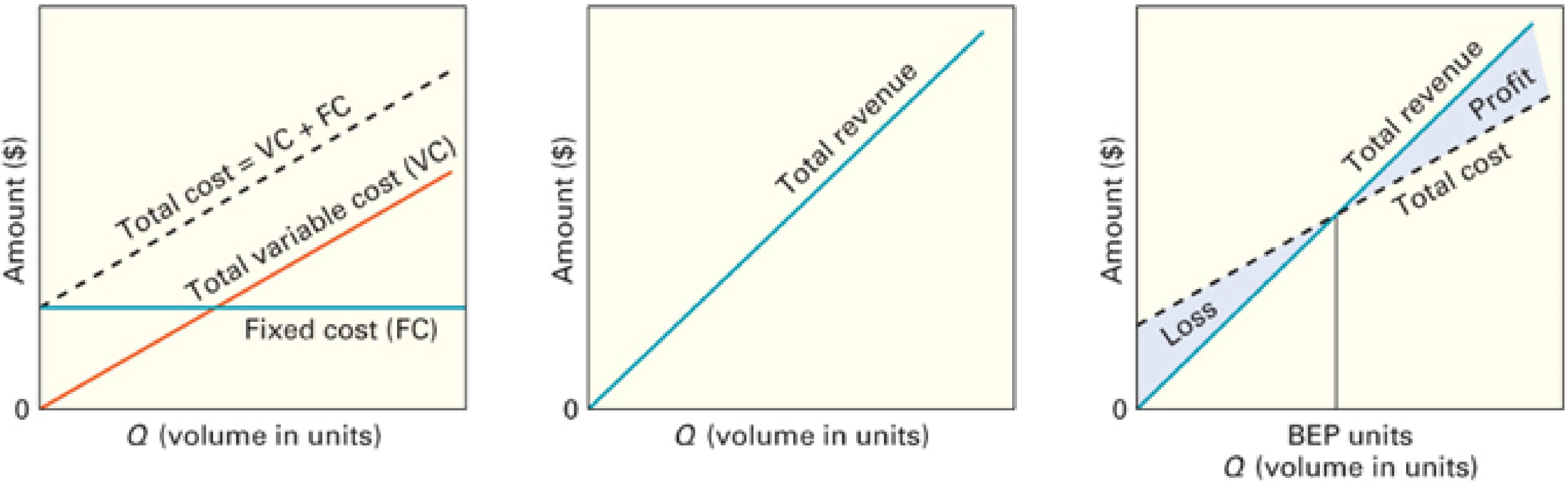
Q = P + FC / (R - v) = 4,000 + 6,000 / (7-2) = 2,000 (pies)

d) ***Profit(P) = Q(R - v) - FC***

=> 5,000 = 2,000 * (R-2) -6,000

=> R = $7.5

Cost-Volume Relationships



A. Fixed, variable, and total costs

B. Total revenue increases linearly with output

C. Profit = TR – TC

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Cost-Volume Analysis Assumptions

- Cost-volume analysis is a viable tool for comparing capacity alternatives if certain assumptions are satisfied.
- **Assumptions:**
    - One product is involved
    - Everything produced can be sold
    - The variable cost per unit is the same regardless of volume
    - The revenue per unit is the same regardless of volume
    - Fixed costs do not change with volume changes
    - Revenue per unit exceeds variable cost per unit

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Financial Analysis

- **Cash flow**
  - The difference between ***cash received*** from sales or other sources and ***cash outflow*** for labor, material, overhead, and taxes.
- **Present value**
  - The total value of all *discounted future cash flows* from the *investment proposal*.
- Financial analysis methods:
  - Payback period
  - Net present value (NPV)
  - Internal rate of return (IRR)

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Payback Period

- Focuses on the *length of time* it will take for an investment to return its original cost.
- More suitable for *short-term* than *long-term* investment.
- ***Equation:***

$$Payback\ period = \frac{Initial\ cost}{Each\ period\ of\ net\ cash\ flow}$$

LO 5.8

Payback Period

- ***Practice:***

  A new machine will cost $2,000, but it will result in savings of $500 per year. What is the payback period in year:

- ***Solution:***

Net Present Value (NPV)

- The difference between the *present value of* ***cash inflows*** and the *present value of* ***cash outflows*** over a *period of time*.
- Summarizes the *initial cost*, estimated *annual cash flows*, and *residual value* in a single value (called the ***Equivalent Current Value***).
- Takes into account the ***time value*** of money (e.g. *interest rates*).

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Net Present Value (NPV)

- ***Equation:***

$$NPV = \sum_{t=1}^{n} \frac{R_t}{(1+i)^t} - R_0$$

where

R_0 = Initial investment cost

R_t = Net cash inflow-outflows during a single period *t*

i = Discount rate or return be earned in investments

t = Number of time period

LO 5.8

Net Present Value (NPV)

If...	It means...
NPV > 0	The investment would add value to the company
NPV < 0	The investment would subtract value from the company
NPV = 0	The investment would neither gain nor lose value for the company

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Net Present Value (NPV)

- ***Practice:***

A corporation invests a new product line with **immediate costs of $100,000**. Assuming this product line will provide equal **benefits of $10,000** for each year, and the effective **annual discount rate is *5%***, please calculating the NPV for the first five years.

Year (*T*)	Cash Flow	Present Value (PV)
0	-100,000	
1		
2		
3		
4		
5		

Net Present Value (NPV)

- ***Solution:***

Year (T)	Cash Flow	Present Value (PV)
0	- $ 100,000	-100,000
1	$ 10,000	$10,000 / (1 + 5\%)^1 = \$\ 9,523.81$
2	$ 10,000	$10,000 / (1 + 5\%)^2 = \$\ 9,070.29$
3	$ 10,000	$10,000 / (1 + 5\%)^3 = \$\ 8,638.38$
4	$ 10,000	$10,000 / (1 + 5\%)^4 = \$\ 8,227.02$
5	$ 10,000	$10,000 / (1 + 5\%)^5 = \$\ 7,835.26$

NPV of first five years =

Internal Rate of Return (IRR)

- Summarizes the *initial cost*, estimated *annual cash flows*, and *residual value* in a single rate (called the ***Equivalent Interest Rate***).
- Identifies the *rate of return* that equal to the estimated future returns and the initial cost.
- Simply put, IRR is the *discount rate* that makes NPV equal zero.

LO 5.8

Internal Rate of Return (IRR)

- ***Equation:***

$$0 = \sum_{t=1}^{n} \frac{R_t}{(1+IRR)^t} - R_0$$

where

R_0 = Initial investment cost

R_t = Net cash inflow-outflows during a single period *t*

IRR =The internal rate or return

t = Number of time period

LO 5.8

Internal Rate of Return (IRR)

- ***Practice:***

A corporation invests a new product line with **immediate costs of $100,000**. Assuming this product line will provide equal **benefits of $10,000** for each year, and this firm wants to pay back in 12 year. Please calculating the IRR.

Year (*T*)	Cash Flow	Present Value (PV)	Year (*T*)	Cash Flow	Present Value (PV)
0	-100,000				
1	10,000		7	10,000	
2	10,000		8	10,000	
3	10,000		9	10,000	
4	10,000		10	10,000	
5	10,000		11	10,000	
6	10,000		12	10,000	

Internal Rate of Return (IRR)

- ***Solution:***

Year (T)	Cash Flow	Present Value (PV)	Year (T)	Cash Flow	Present Value (PV)
0	-100,000				
1	10,000	9,716.02	7	10,000	8,173.67
2	10,000	9,440.09	8	10,000	7,941.55
3	10,000	9,172.01	9	10,000	7,716.02
4	10,000	8,911.54	10	10,000	7,496.90
5	10,000	8,658.46	11	10,000	7,284.00
6	10,000	8,412.58	12	10,000	7,077.15

$0 = \$10{,}000 / (1+IRR)^1 + \$10{,}000 / (1+IRR)^2 + \ldots + \$10{,}000 / (1+IRR)^{12} - \$100{,}000$

***IRR* = 0.0292 = 2.92%**

Operations Strategy

Capacity planning impacts all areas of the organization

- It determines the conditions under which operations will have to function
- *Flexibility* allows an organization to be agile
  - It reduces the organization's dependence on forecast accuracy and reliability
  - Many organizations utilize capacity cushions to achieve flexibility
- *Bottleneck management* is one way by which organizations can enhance their effective capacities
- *Capacity expansion* strategies are important organizational considerations
  - Expand-early strategy
  - Wait-and-see strategy
- *Capacity contraction* is sometimes necessary
  - Capacity disposal strategies become important under these conditions

Summary

- Capacity refers to a system's potential for producing goods or delivering services over a specified time interval. Capacity decisions are important because capacity is a ceiling on output and a major determinant of operating costs.
- Three key inputs to capacity planning are the kind of capacity that will be needed, how much will be needed, and when it will be needed. Accurate forecasts are critical to the planning process. The capacity planning decision is one of the most important decisions that managers make.

Summary

- The capacity decision is strategic and long-term in nature, often involving a significant initial investment of capital. Capacity planning is particularly difficult in cases where returns will accrue over a lengthy period and risk is a major consideration.
- A variety of factors can interfere with effective capacity, so effective capacity is usually somewhat less than design capacity. These factors include facilities design and layout, human factors, product/ service design, equipment failures, scheduling problems, and quality considerations.

Summary

- Capacity planning involves long-term and short-term considerations. Long-term considerations relate to the overall level of capacity; short-term considerations relate to variations in capacity requirements due to seasonal, random, and irregular fluctuations in demand. Ideally, capacity will match demand.
- Thus, there is a close link between forecasting and capacity planning, particularly in the long term. In the short term, emphasis shifts to describing and coping with variations in demand.

Summary

- Development of capacity alternatives is enhanced by taking a systems approach to planning, by recognizing that capacity increments are often acquired in chunks, by designing flexible systems, and by considering product/service complements as a way of dealing with various patterns of demand.
- In evaluating capacity alternatives, a manager must consider both quantitative and qualitative aspects. Quantitative analysis usually reflects economic factors, and qualitative considerations include intangibles such as public opinion and personal preferences of managers. Cost–volume analysis can be useful for analyzing alternatives.

The End